INTERNAL · FORENSIC · INCONCLUSIVE (UNPROFILEABLE IMAGE) · 1 APPROVED FINDING

AX AGENTROPIX-SIFT · DFIR

FULL PROFILE · REAL-DATA RUN · UNPROFILEABLE MEMORY IMAGE

CTF contact_me — Memory-Triage Forensic Report

Single 1 GiB raw RAM image · DFIR memory triage · outcome: UNPROFILEABLE / inconclusive

REPORT ID	54bc98b34355314f029aae4a4a69d2bc3a93fccbae80d44da6ae6bed6cbbce8a
CASE ID	CTF-CONTACT-ME-MEM
CASE NAME	CTF contact_me (raw memory)
EXAMINER	victor.galvan
PROFILE	full
INCIDENT TYPE	dfir
SEVERITY	medium
SNAPSHOT	2026-06-07T04:05:02.339181+00:00
GENERATED	2026-06-07T04:05:02Z
MCP HOST	

APPROVED · approval_id 837a4ad5f033953b2cbcbd2d5baf9daf9c60993bce449089dc776a9b131bc7cc · examiner victor.galvan · approved 2026-06-06T23:17:38Z (SIMULATED examiner sign-off — demo)

01 · EXECUTIVE DASHBOARD

Executive Summary

This report covers a single 1 GiB raw RAM capture (`/cases/contact_me/contact_me`) triaged through the Agentropix-SIFT MCP memory pipeline. The headline result is honestly negative: **the image is unprofileable.** Volatility3 2.28.0 could not validate a Windows kernel symbol table for this capture, so every windows.* plugin (pslist / netscan / malfind / svcscan) returned placeholder or empty output. No processes, sockets, services, injected code, IOCs, or timeline events could be resolved. There is therefore **no clean-vs-compromised determination** to make from this evidence. One finding (F-CONTACTME-001, medium) was recorded and approved to document this unprofileable outcome through the full init → evidence → finding → approval → sealed-report loop.

Key finding — INCONCLUSIVE. The capture is not profileable by Volatility3 2.28.0: the kernel layer / symbol table could not be validated, so analysis tools yield no substantive artefacts. A populated process list — not an HTTP-200 health check — is the true signal that a memory profile matched; that signal is absent here. Re-acquisition or a matching symbol table is required before any conclusion.

1
APPROVED FINDINGS
`medium: 1 (documents the unprofileable outcome)`

1
HOSTS IN SCOPE
`single RAM image (host: contact_me)`

0
IOCS CATALOGUED
`none catalogued — image unprofileable`

N/A
DWELL / WINDOW
`undeterminable from RAM alone (no boot/timeline anchor)`

0
ATT&CK TECHNIQUES
`no MITRE techniques observable`

—
INITIAL ACCESS
`inconclusive (no resolvable processes/timeline)`

Hosts / evidence roster

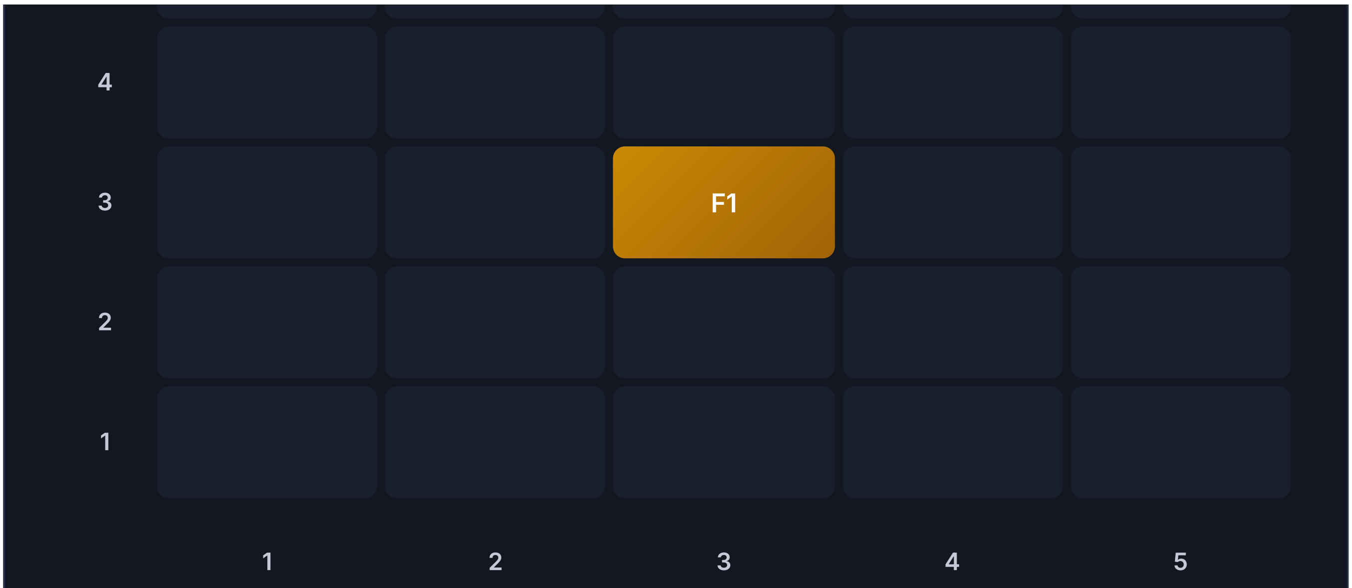
🧠
`contact_me`
1 GiB raw RAM image · unprofileable

02 · RISK ASSESSMENT

Risk Matrix

Because the image is unprofileable, there is no technical evidence on which to score adversary risk. The single approved finding is scored only as a **process / coverage risk** (PROJECTION): moderate impact (a triage tool could not produce results, leaving a blind spot) at moderate likelihood (symbol-table mismatches are a common, recurring acquisition issue). This is a methodology score, not a compromise score — no compromise has been established or ruled out. *Scoring is qualitative (operator-derived from approved-finding severity + technique), not a tool-computed number.*

5



X-axis: Likelihood (1-5) · Y-axis: Impact (1-5) · Risk = Impact × Likelihood

REF	RISK	IMPACT	LIKELIHOOD	SCORE	SEVERITY
F1	Unprofileable memory image — triage blind spot (no compromise determination)	3 (Moderate)	3 (Possible)	9	MEDIUM

03 · KEY FINDINGS

Key Findings

MEDIUM F-CONTACTME-001

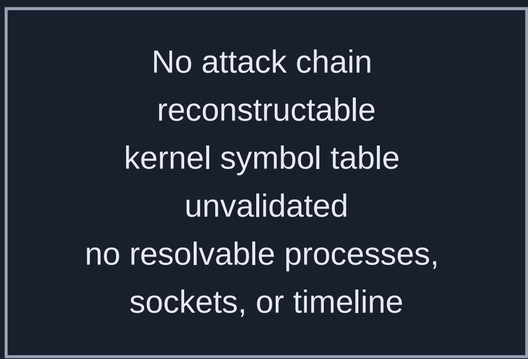
Memory image unprofileable — no clean-or-compromised determination possible

Volatility3 2.28.0 could not validate a Windows kernel symbol table for this capture (`plugins.PsList.kernel.symbol_table_name` / `kernel.layer_name` requirement validation failed). As a result `pslist` returned 11 placeholder rows (all `pid:0 name:'unknown'`), `netscan` 0 sockets, `malfind` 0 hits, `svcscan` 0 services, `build_process_tree` a single empty root, and `cmdline` a non-JSON requirements error. These are **degraded/gated results, not confirmed-clean results**. Analytic confidence **high** (LCA) that the image is unprofileable asacquired; likelihood *unlikely*; risk score 6; provenance MCP; approved 2026-06-06 (simulated examiner sign-off for the demo).

04 · ADVERSARY

Attack Chain & MITRE ATT&CK

No attack chain is reconstructable. With no validated kernel symbol table there are no resolvable processes, parent/child relationships, network sockets, injected-code regions, services, or timeline anchors — so no kill-chain phase, lateral movement, or persistence can be observed or excluded. The kill-chain diagram below is therefore an explicit **inconclusive** marker, not a derived chain. No MITRE ATT&CK techniques were observed.



Kill-chain / attack progression (flowchart)

MITRE ATT&CK techniques observed

N/A

—

None observed

No techniques are observable from an unprofileable image. The approved finding carries an empty `mitre_attack` field by design — fabricating a technique would violate the grounding contract.

05 · INDICATORS

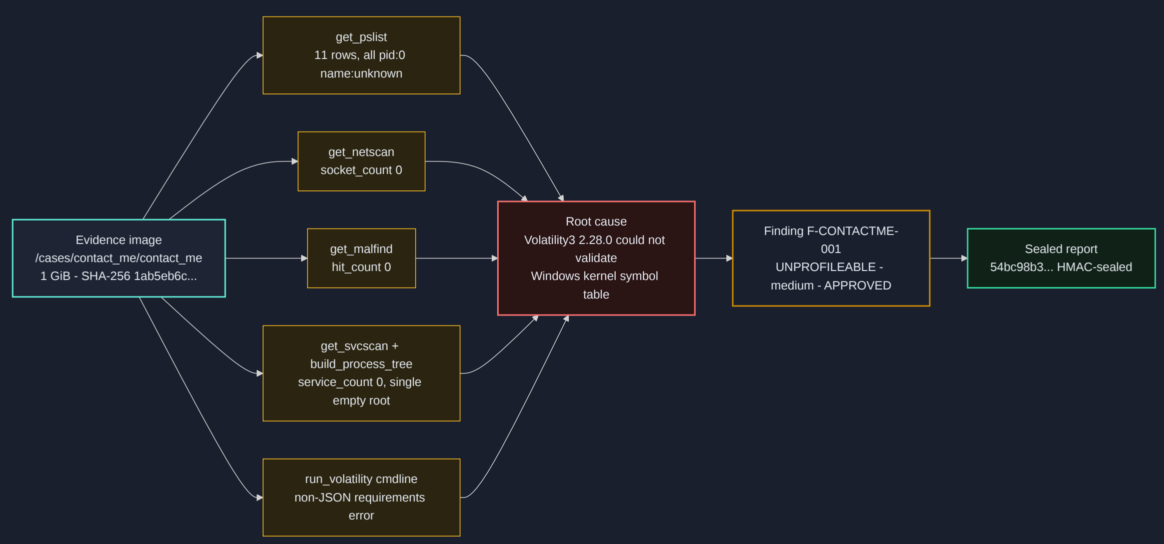
IOC Catalogue

No IOCs were catalogued. The `report_generate(profile=full) iocs` section returned `count: 0` (no by-type, no by-MITRE entries). No network, file, or identity indicators could be surfaced because the underlying plugins could not resolve any process or kernel structures from the image.

TYPE	VALUE	ROLE	CONFIDENCE	MITRE	SOURCE
No network/file/identity IOCs surfaced in this memory image — the capture is unprofileable.					

> **Negative space (surveyed, none confirmed):** IPv4/IPv6 endpoints (netscan → 0 sockets), process command lines (cmdline → requirements error), injected/RWX code regions (malfind → 0 hits), service binaries (svcscan → 0 services), and process ancestry (process tree → single empty root). Every category was queried against real plugins; none returned data because the symbol table did not match.

IOC provenance



`IOC → evidence-source provenance graph`

06 · HOST ARTEFACTS

Host Artefacts

Single memory image; no on-disk artefacts in scope. All host-artefact tables below reflect the degraded plugin output verbatim — placeholder rows from pslist and empty service/socket results — with the verbatim Volatility requirements error in the Source column so the reason is auditable.

Host: contact_me (1 GiB raw RAM)

PID	PPID	IMAGE	STARTED	STATE	WOW64	NOTES	SOURCE
0	0	unknown	—	—	—	Placeholder row — kernel symbol table unvalidated; process not resolved	get_pslist · process_count=11, all rows pid:0/unknown · raw_stderr: "Unable to validate the plugin requirements: [plugins.PsList.kernel.layer_name, plugins.PsList.kernel.symbol_table_name]"

Remaining 10 rows identical placeholders (pid:0 / name:"unknown"). No process is genuinely identifiable.

Services (svcscan)

SERVICE	IMAGE PATH	INSTALL UTC	CLASS	SOURCE

service_count=0 — volatility3.windows.svcscan.SvcScan returned no services (same unvalidated-symbol-table gate). Not a confirmed "no services" result.

malfind (RWX / injected code): hit_count=0 from volatility3.windows.malfind.Malfind — gated by the same requirements error, so this is **not** a confirmed-clean RWX scan.

Process tree

Single empty root - pid:0 name:unknown build_process_tree: process_count 11, root_count 1, orphan_count 0, suspicious_count 0 (placeholder rows, no valid ancestry)

Process ancestry (build_process_tree)

07 · NETWORK ARTEFACTS

Network Artefacts

No network connections could be enumerated. `get_netscan` returned `socket_count: 0` from volatility3.windows.netscan.NetScan — but for the same reason as the other plugins (no validated kernel layer), so this is not a clean "no connections" finding.

PROCESS (PID)	LOCAL	REMOTE	STATE	PURPOSE	SOURCE
netscan returned 0 sockets — gated by the unvalidated symbol table, not a confirmed empty network state.					

raw_stderr (verbatim): `Unable to validate the plugin requirements: [plugins.NetScan.kernel.layer_name, plugins.NetScan.kernel.symbol_table_name]`

08 · DETAILED FINDINGS

Detailed Findings & Evidence

MEDIUM F-CONTACTME-001

Memory image unprofileable: Volatility3 2.28.0 could not validate a Windows kernel symbol table

pslist/netscan/malfind/svcscan returned placeholder/empty results; no clean-or-compromised determination possible. host=contact_me · severity=medium · likelihood=unlikely · confidence=high (LCA) · risk_score=6 · mitre_attack=(empty) · provenance=MCP.

EVIDENCE

- evidence SHA-256 `1ab5eb6c3b87a0604f75a00cb4a64d91aaf7ab4e303bb7337b40f7e3df8ad61a` (/cases/contact_me/contact_me, 1073741824 bytes = 1 GiB)
- get_pslist: process_count=11, all rows placeholder `pid:0 name:'unknown'`; raw_stderr `'Unable to validate the plugin requirements: [plugins.PsList.kernel.layer_name, plugins.PsList.kernel.symbol_table_name]'`
- get_netscan: socket_count=0 (volatility3.windows.netscan.NetScan)
- get_malfind: hit_count=0 (volatility3.windows.malfind.Malfind)
- get_svcscan: service_count=0; build_process_tree: process_count=11 root_count=1 single pid:0 'unknown' root, suspicious_count=0
- run_volatility plugin=cmdline: error `'vol3 emitted non-JSON output: Expecting value: line 2 column 1 (char 1)'`

APPROVAL `approval_id:837a4ad5f033953b2cbcbd2d5baf9daf9c60993bce449089dc776a9b131bc7cc` (sha-256 approval record · approved 2026-06-06T23:17:38Z · SIMULATED examiner sign-off)

09 · TIMELINE

Timeline (UTC)

TIME (UTC)	HOST	EVENT	TECHNIQUE	SOURCE
No temporally-anchored events; the report_generate timeline section returned count: 0 and the RAM snapshot yields no resolvable timestamps.				

No timeline. The sealed `timeline` section is empty (`approved_timeline_events: [], count: 0`). Without resolvable processes or kernel structures there are no creation/connection events to anchor, and a single RAM snapshot carries no intrinsic event sequence.

Agentropix Performance

The run executed the full manual triage sequence end-to-end: environment doctor, health, case init/activate/status, evidence register (real SHA-256), the four windows.* plugins, process-tree correlation, cmdline, a dry-run finding, a real indexed finding, simulated examiner approval, and a sealed report. Every MCP call returned (HTTP 200 / status ok); the analysis plugins ran but were gated by the symbol-table mismatch. Stage-level wall-clock timings were not separately instrumented in this run, so per-stage durations are reported as not-captured rather than fabricated.

N/C
TOTAL RUNTIME
`per-stage timings not instrumented this run`

13
MCP TOOL CALLS
`MCP steps in EXECUTED-RUN (doctor→sealed report)`

1 GiB
EVIDENCE PROCESSED
`1,073,741,824 bytes (exact)`

N/A
RECALL
`recall undefined — image unprofileable`

STAGE / TOOL	CAPTURE	DURATION	RESULT
doctor / health	tool inventory + tool_count	N/C	All tools available · tool_count 72 (live)
case_init / activate / status	CTF-CONTACT-ME-MEM	N/C	active:true, indexer_reachable:true
evidence_register	/cases/contact_me/contact_me	N/C	SHA-256 recorded, 1 GiB, indexed:true
get_pslist	process list	N/C	status ok · 11 placeholder rows (symbol table unvalidated)
get_netscan	sockets	N/C	socket_count 0 (gated)
get_malfind	RWX / injected code	N/C	hit_count 0 (gated)
get_svcscan / build_process_tree	services + ancestry	N/C	service_count 0 · single empty root

STAGE / TOOL	CAPTURE	DURATION	RESULT
run_volatility cmdline	command lines	N/C	non-JSON requirements error (verbatim)
record_finding (dry_run)	ctf-contactme-001	N/C	validated DRAFT (indexed:false)
record_finding (real)	F-CONTACTME-001	N/C	indexed:true
examiner approval (simulated)	F-CONTACTME-001	N/C	DRAFT → APPROVED, HMAC challenge-response
report_generate (profile=full)	sealed report	N/C	approved_finding_count 1, report_id 54bc98b3...

11 · ATTESTATION

Coverage Attestation

Coverage here is attested at the **pipeline** level, not the artefact level: the full tool chain ran and the evidence-gated, approval-only sealing path was respected. The platform-wide recall figures below are canonical regression numbers (per .crew/facts.md) and are NOT a claim about this image — recall is undefined for an unprofileable capture.

Disk recall **72/72 (100%)**	Memory recall **108/118 (91.5%)**
Tests passing **4464**	Approval gate **APPROVED-only**

> This report contains exactly the 1 APPROVED finding for case CTF-CONTACT-ME-MEM and 0 unapproved items — the EvidenceGate / approval-only path was honoured (DRAFT findings are excluded from sealed output). The report is HMAC-sealed (report engine, ADR-024). The single finding faithfully records an unprofileable outcome rather than padding the report with unsupported claims.

12 · RECOMMENDATIONS

Recommendations

P1 Re-acquire or re-profile: confirm the OS/build of the source host and supply a matching Volatility3 symbol table (ISF) for it, or re-acquire the memory with a tool that records the kernel build, then re-run pslist/netscan/malfind/svcscan. A populated process list — not an HTTP-200 — is the signal a profile matched.

P2 Verify image type and integrity: confirm the capture is a genuine Windows physical-memory image (not a partial/compressed/encrypted dump) and re-hash against the recorded SHA-256 (`1ab5eb6c3b87a060…`) before re-analysis.

P3 Do not treat the 0-count results as a clean bill of health: netscan/malfind/svcscan returning 0 is a tool-gated artifact here, not a forensic conclusion. Hold the case open as INCONCLUSIVE pending a successful profile match.

13 · APPENDIX

Appendix

Methodology

Manual triage sequence (3A) executed against the live Agentropix-SIFT MCP server: doctor/health → case_init/activate/status → evidence_register → the four windows.* analysis plugins (pslist, netscan, malfind, svcscan) → build_process_tree correlation → run_volatility(cmdline) → record_finding (dry-run then real, evidence-gated) → examiner approval (simulated for the demo) → report_generate(profile=full). The report engine (ADR-024) projects only APPROVED findings into HMAC-sealed sections. All values in this document come from the sealed report_generate output and the case EXECUTED-RUN captures; nothing is inferred or fabricated.

Tool & framework versions

`Python`

3.12+

`Volatility 3 Framework`

2.28.0 (could not validate a Windows kernel symbol table for this image)

Agentropix-SIFT MCP

server 0.1.0-dev · live tool_count 72 (canonical 71)

Report engine

ADR-024 · HMAC-SHA256 sealing

Mermaid (diagrams)

mermaid-cli bundled mermaid.min.js

Tests (platform)

4464 passing

Chain-of-custody hashes

ARTEFACT	ALGO	DIGEST	STATUS
contact_me memory image (/cases/contact_me/contact_me, 1 GiB)	SHA-256	1ab5eb6c3b87a0604f75a00cb4a64d91aaf7ab4e303bb7337b40f7e3df8ad61a	RECORDED
evidence_id (custody record)	SHA-256	6d9dcf5ffe92f6da401d60745402ba19c42d4db7a48ee6ffb27bd461bbb4f142	REGISTERED
approval record (F-CONTACTME-001)	SHA-256	837a4ad5f033953b2cbcbd2d5baf9daf9c60993bce449089dc776a9b131bc7cc	APPROVED

Provenance: all artefacts trace to a single registered evidence image (SHA-256 above) processed through the Agentropix-SIFT MCP at host . The unprofileable outcome is recorded honestly — the absence of processes, sockets, IOCs, and timeline events reflects a tool-gating symbol-table mismatch, not a clean system. No facts were fabricated to fill empty sections.
